A novel case of RAG1 (Recombination Activating Gene) deficiency presenting with immunodeficiency in adulthood

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Mutations in the recombination activating genes (RAG) have been reported to cause severe combined immunodeficiency (SCID). Hypomorphic mutations may result oligoclonal T cells responsible for an Omenn syndrome phenotype. RAG1 mutations resulting in immunodeficiency generally present at an early age. The increased use of whole exome and whole genome sequencing methods has resulted in deeper investigation of primary immunodeficiencies (PID). An increasing number of adult PIDs are now being identified. We present a case of biallelic missense mutations in RAG1 presenting with immunodeficiency in adulthood.

Immunological assessment

NGS
Clinical panel
of PID genes

RAG1

c.1210C>T (Arg404Trp)
c.1520G>A (Arg507Gln)

Murine protein
R401W
R504Q

Clinical Features

Immunological assessments

Tests	Values	Normal ranges
IgG (g/l)	8.5	5.4-16.1
IgA (g/l)	3.31	0.8-2.80
IgM (g/l)	0.37	0.5-1.90
IgE ku/l	<2.0	0.5-120
Pneumococcal Ab µg/ml	64.9	Adequate >30
Tetanus IU/ml	1.614	Adequate >0.15
Haemophilus µg/ml	>9000	Adequate > 1
Total lymphocytes (cells/µl)	1248	1000-2800
CD3+ Lymphocytes (cells/µl)	926	800-3500
CD4+ T cells (cells/µl)	280	300-1400
Naïve CD4+ T cells (%)	26	>40
T reg CD+ T cells		>0.5
CD8+ T cells (cells/µl)	526	200-900 cells/µl
NK cells CD56+ (cells/µ)l	210	90-600 cells/µl
B cells (CD19+) cells/µl	31	100-500 cells/µl
ratio	0.53	1.07-1.87
Marginal zone B cells CD19+ CD27+ IgD+ (% of CD19+)	3	0.5-8%*
Class switched memory B cells CD19+ CD27+ IgD- (% of CD19+)	58	3-18%*
PHA induced lymphocyte proliferation	Reduced	N/A
Anti-CD3 induced lymphocyte proliferation	Preserved	N/A
Autoimmune screen [ANA, AMA, ACPA, Endomysial Ab (IgG and IgA)]	Negative	N/A

MW-Male age 33

Childhood history of 'asthma' and recurrent ear infection (repeated grommets insertions)

Severe & recurrent viral warts (hands and feet) age 6 (significantly better after age 20)

Oral candida and fungal infections in later life

Diagnosed with bronchiectasis age 26 after repeated episodes of chest infections

One episode of reactive arthritis after following pneumonia

No other autoimmune/inflammatory complications

No family history

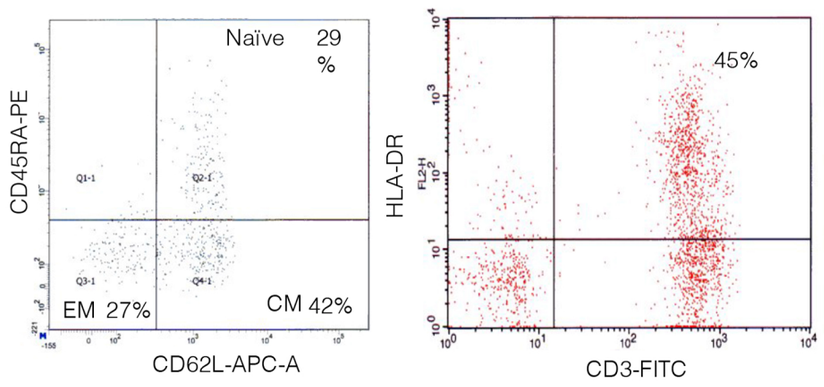
In vivo T cell activity

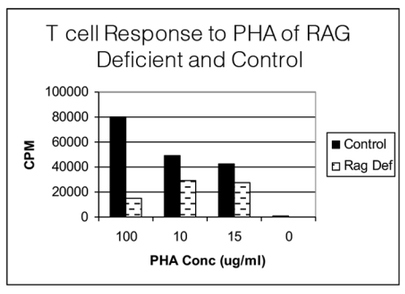
Immunological assessment indicates moderate CD4+ and severe B cell lymphopenia. Immunoglobulin levels were within the normal range. Antibody function also appeared normal.

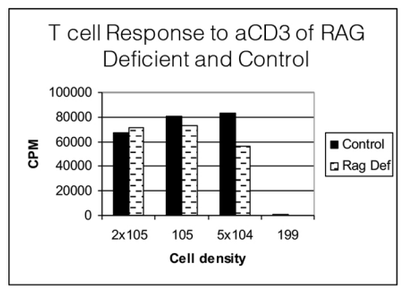
Most cases of SCID/Omenn syndrome incur a skewing towards a naïve T cell population. However, the patient showed memory T cells present at 42% (central memory) and 27% (effector memory). T cell activation was recorded with impaired T cell proliferation.

Flow cytometric T-cell receptor Vβ repertoire analysis is a sensitive method for detection of T-cell clonality. Reductions were seen in TCR Vβ 2 and 23 along with a large number of oligoclonal Vβ 14 T cells.

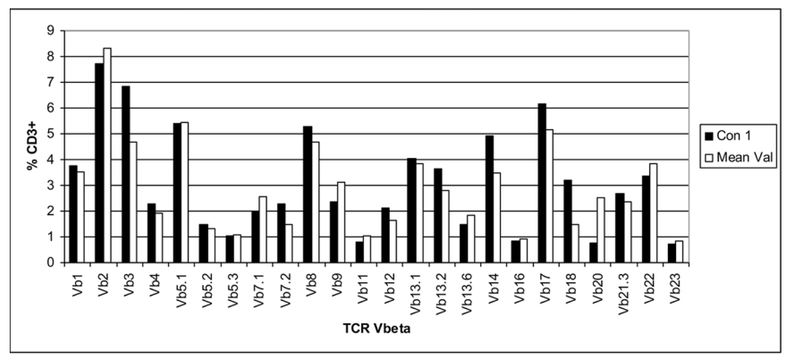
CD4 naïve/memory/activated and T cell proliferative responses



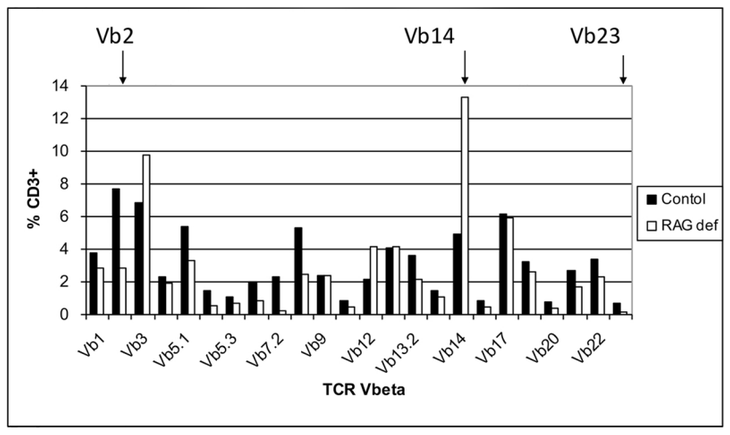




TCR Vbeta in healthy control

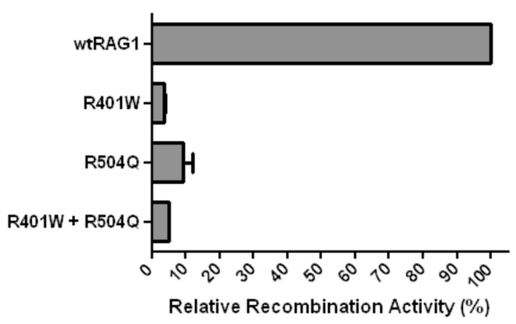


TCR Vbeta in patient



Protein function

Recombination Activity



To assess the effect of mutation on protein function murine RAG1 contructs were produced. An extrachromosomal recombination assay was used based on RAG protein recognition of 12 and 23 nucleotide recombination signal sequences (RSSs) during V(D)J recombination. Both variants show loss of function.

Conclusion

With the increase of high accuracy NGS methods the identification of genetic determinants is becoming more frequent for adult PID. It is possible to investigate subtle protein functions through discovery of diseases with untypical presentation.

We report a case of biallelic mutations in RAG1 resulting in adequate immunoglobulin productions and altered T cell activity. Identification of such cases are expected more frequently in the future.

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